

Validation and authentication

Revise 360 lesson plan — 60 minutes

Unit	Beyond OCR J277: a Python pathway supporting 2.1, 2.2 and 2.3
Lesson	Validation and authentication (lesson 12 of the unit)
Lesson ID	pr-l12
Experience ID	pr-l12
Plan ID	PLAN-PR-L12
Content version	162820f82a12
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Big question, outcomes and success criteria

Big question — Why is refusing bad data better than letting a program stop with an error?

Prerequisites

The lesson before this one is pr-l11. Pupils should already have finished it: the starter below retrieves from it.
Earlier in the unit: pr-l01, pr-l02, pr-l03, pr-l04, pr-l05, pr-l06, pr-l07, pr-l08, pr-l09, pr-l10.

Assessable outcomes

- pr-l12-o1** — know how to code a presence check, a type check and a range check
- pr-l12-o2** — understand how a length check and a format check differ, and why the length is tested first
- pr-l12-o3** — know how to write a validation loop, and an authentication check that allows a set number of attempts

Success criteria the pupil can check against

Taken from the briefs the independent activities themselves show, so these are the words the pupil sees. Their work has met the outcome when it does these things.

- Display exactly: nothing typed pr-l12-s1-a3
- The user types in one line. pr-l12-s1-a4
- Display exactly: no colour given pr-l12-s1-a4
- One line only is displayed. pr-l12-s1-a4
- Change only the two comparisons. pr-l12-s2-a3
- Display exactly: valid or exactly: refused pr-l12-s2-a3
- Every line typed is a whole number. pr-l12-s2-a4
- Both 1 and 6 are allowed. pr-l12-s2-a4

Vocabulary, resources and preparation

Key vocabulary — Validation, Format check, Validation loop, Authentication

Resources

SLIDES	presentations/PR_L12_ValidationAndAuthentication_Lesson.pptx
WORKSHEET	worksheets/PR_L12_ValidationAndAuthentication_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pr-l12

Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the Python runtime and the 360 image are already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>PR_L12_ValidationAndAuthentication_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong before moving on.	Correct their own starter.	A correction written, not just a tick.
8-16	Explain and model	The new idea, then the worked example, typed live rather than shown finished.	Watch the example being built, and say what each line does.	They can name the line that does the job, not just copy it.
16-19	Set up and settle	Sign-in, launch, and what to record on paper before anyone types.	Sign in, open the lesson, read the task before typing.	Everyone is at station 1 activity 1, not typing yet.
19-48	Online: work the stations in order	Pupils work; you circulate and read the editor over shoulders.	Work the stations in order, running and checking as they go.	The station badges fill in; the editor shows their own program, not the starter.
48-53	Check understanding	The hinge question, hands down, then the reteach if the class splits.	Answer the hinge question, hands down.	The split across the options, before any reteach.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
53-58	Apply or extend	The challenge activity, or the exam practice on the worksheet.	The Challenge activity, or the exam practice.	worksheet, Exam practice a) to c)
58-60	Exit and set up the next lesson	The exit question and what to finish before next time.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Checking the input

- A presence check refuses input that is empty once any stray spaces are stripped off.
- A type check uses `isdigit()`, so text typed where a number was wanted cannot crash the program.
- Common mistake: casting with `int()` before checking, which stops the program instead of rejecting the value.

Challenge on this wall: Why must a type check happen before the value is cast?

2. Range checks

- A range check refuses anything that falls outside the allowed limits.
- Inclusive limits need `>=` and `<=`, so the boundary values themselves are accepted.
- Common mistake: using `>` where the rule says 'or more', which rejects a perfectly valid boundary.

Challenge on this wall: Which two values would you test on a score that must be 1 to 10?

3. Length and format checks

- A length check counts the characters with `len()`, against a minimum and a maximum.
- A format check tests the pattern: which positions must be letters and which must be digits.
- Common mistake: checking the pattern before the length, which can read past the end of the string.

Challenge on this wall: Describe the checks a code of two letters then four digits needs.

4. Keeping asking

- A validation loop reads once, then repeats for as long as the value breaks a rule.
- Saying which rule was broken makes the program far easier to use.
- Common mistake: leaving the second `input()` out, so the loop tests the same bad value forever.

Challenge on this wall: Why should a program say what was wrong rather than just 'invalid'?

5. Authentication

- Authentication checks who the user is, usually with a username and a password.
- An attempt counter lets a program lock the account after a set number of tries.
- Common mistake: counting the attempts but never stopping, so guessing can carry on forever.

Challenge on this wall: What is the difference between validation and authentication?

6. Putting it together

- Validate each value in its own loop before the program moves on to the next one.
- A program that keeps going when one line of data is wrong is more useful than one that stops.
- Common mistake: validating at the end, when the first bad value has already been used.

Challenge on this wall: Which checks does a username need, and in which order should they run?

The example to model

Validation means checking what was typed before the program uses it. A presence check is the simplest one: it asks whether anything was typed at all. `input()` hands back an empty piece of text when Enter is pressed alone. Two quotation marks with nothing between them, `""`, are how that empty text is written.

```
entry = input("Name? ")    # the user pressed Enter straight away
if entry == "":
    print("nothing typed")
else:
    print("thank you")
```

It displays **nothing typed**

Decisions to model out loud, not just the finished answer: say why you chose that name, why that line comes before the next one, and what you would try first if it did not work. The example deliberately uses different data from the task, so a pupil cannot copy it and be finished.

Retrieval and the check for understanding

Starter — A program reads an age with `age = int(input())` and the user types the word twelve. Write down what happens to the program, and the check that should have come before the cast.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pr-112-s4-c2), so the wording a pupil sees matches the wording you say.

The user types 99 and then 5. What will this program display?

Correct: try again done

What a pupil might choose instead

- done
- try again try again done

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: One wrong line gives one message. The loop checks the new value before it decides whether to go round again.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Misconceptions this lesson addresses

- casting with `int()` before checking, which stops the program instead of rejecting the value. [wall panel](#)
- The check happens on line 3, but something has already been cast on line 2. [pr-112-s1-a8, a Fix it activity](#)
- using `>` where the rule says 'or more', which rejects a perfectly valid boundary. [wall panel](#)
- checking the pattern before the length, which can read past the end of the string. [wall panel](#)
- Nothing in this condition says how long the code has to be. [pr-112-s3-a7, a Fix it activity](#)
- leaving the second `input()` out, so the loop tests the same bad value forever. [wall panel](#)
- An if asks once. What asks again and again until the answer changes? [pr-112-s4-a5, a Fix it activity](#)
- counting the attempts but never stopping, so guessing can carry on forever. [wall panel](#)

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	18	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	8	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Produce **14** The pupil writes it from the brief and nothing else.

Challenge **6** The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Checking the input	9	Validation can only check that data is sensible, never that it is...
2	Range checks	7	A negative value is sensible for a temperature but never for an age, so the limits of a range check come from...
3	Length and format checks	8	isalpha() and isdigit() ask their question of every character at once, so either one is False as soon as one character...
4	Keeping asking	7	A loop that reads, checks and asks again until the value passes is the simplest defensive design there is, and it is called a...
5	Authentication	7	Validation checks the data that was typed, while authentication checks the...
6	Putting it together	7	Naming a bad line and carrying on is usually better than stopping, because the run still...
final	Final challenge: which check?	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Python: the order is not negotiable. Each activity teaches what the next one needs, so a pupil works through them in order and completes every one. A shorter lesson, homework or a slow worker does not authorise skipping: the pupil stops where they are and resumes at their earliest unfinished question next time, because progress saves after every answer.

A paper activity, a help request or a tick on this plan does not mark an online question complete, and there is no teacher bypass. If a pupil is stuck after the hint, that is the moment to teach them, not to move them on.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pr-112-x1	A range check for a score that must be from 1 to 10 inclusive is written <code>if score > 1 and score < 10</code> . State the two scores this wrongly refuses, and write the condition corrected.	3	worksheet, Exam practice a); deck, Exam practice slide
pr-112-x2	Write a program that keeps asking for a code of exactly one letter followed by two digits, saying what is wrong each time, until a valid code is typed.	6	worksheet, Exam practice b); deck, Exam practice slide
pr-112-x3	Write a program that allows three attempts at a four-digit passcode already held in the program, and prints Locked when all three attempts are wrong.	5	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pr-112.json`, which is not published with the website.

Every programming activity in this lesson is marked by running the pupil's program against its test cases, so the mark is the program's behaviour and not its appearance. The model solutions are in `answers/codebank/pr-112.json`.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What will this program display?** — True True; False False
- **What will this program display?** — True True True False; True False True False
- **What will this program display?** — valid; 10
- **What will this program display?** — 6 True True; 6 True False
- **The user types 99 and then 5. What will this program display?** — done; try again try again done

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Coding a presence, type and range check

- Coding a length check and a format check
- Writing a validation loop and an authentication check

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pr-113, SQL queries. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

If this lesson takes more than an hour

It will. There are 46 activities across 6 stations, and a beginner does not finish 46 of them in twenty-nine minutes. Plan it as two or three sessions of an hour: stations 1 and 2 in the first, 3 and 4 in the second, 5, 6 and the final challenge in the third. Each session keeps the same starter-explain-model opening; the online block resumes where the class got to. Honest total: two to three hours.

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

A shorter lesson still does not authorise skipping a question. The save-and-resume point is wherever the pupil got to; there is no way to mark a question complete without completing it, and no teacher bypass.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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